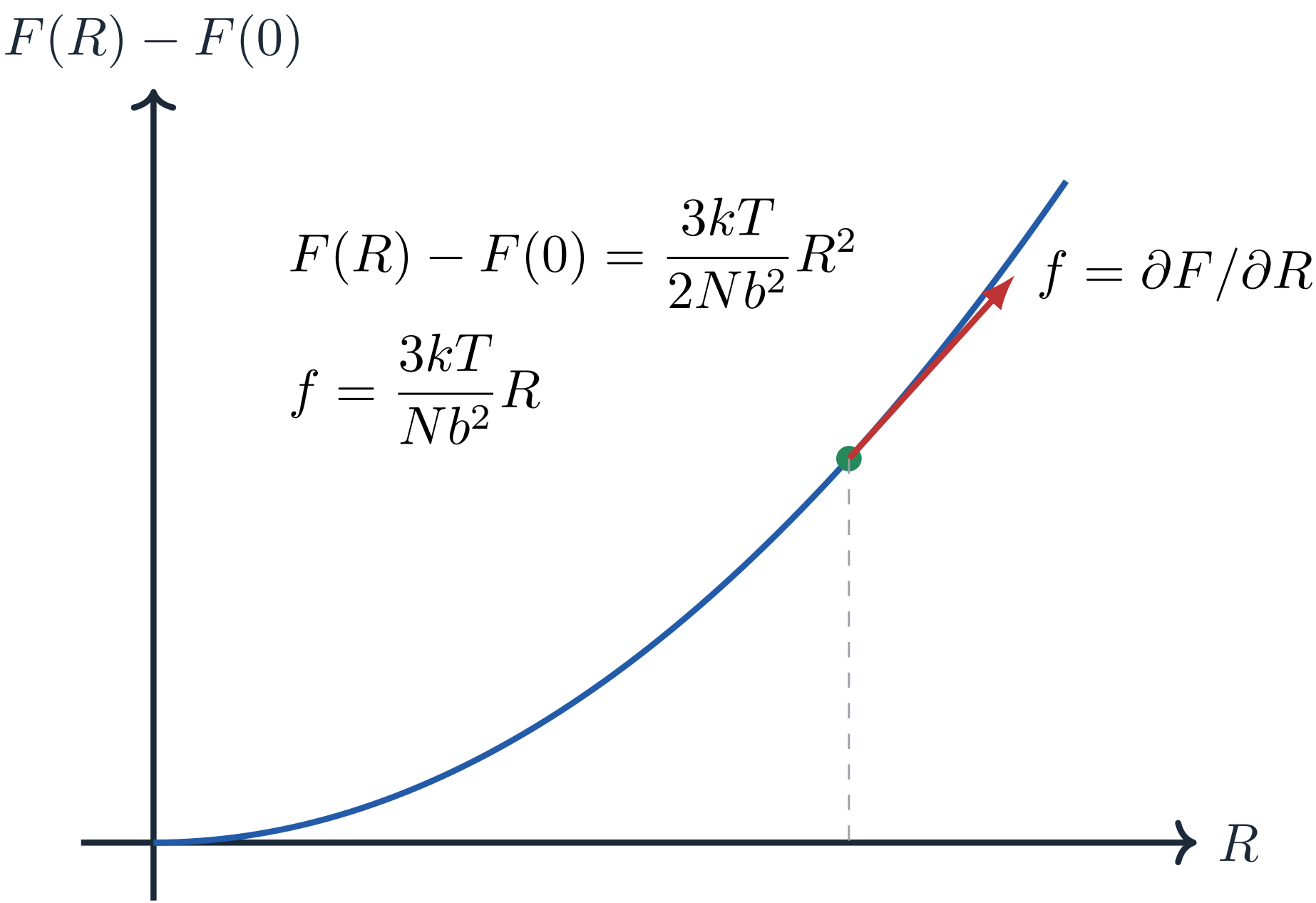
$F(R) - F(0)$
$F(R) - F(0) = \frac{3kT}{2Nb^2} R^2$
$f = \frac{3kT}{Nb^2} R$
$f = \partial F / \partial R$
R